

Linear Independent

The Definition of Linear Independence

Definition. A set of vectors $\{v_1, v_2, \dots, v_k\}$ is **linearly independent** if the vector equation

$$x_1 v_1 + x_2 v_2 + \dots + x_k v_k = 0$$

has **only the trivial solution** $x_1 = x_2 = \dots = x_k = 0$. The set $\{v_1, v_2, \dots, v_k\}$ is **linearly dependent** otherwise.

In other words, $\{v_1, v_2, \dots, v_k\}$ is linearly dependent if there exist numbers $x_1, x_2, \dots, x_k$, not all equal to zero, such that

$$x_1 v_1 + x_2 v_2 + \dots + x_k v_k = 0.$$

This is called a **linear dependence relation** or **equation of linear dependence**.

Theorem. Let $v_1, v_2, \dots, v_k$ be vectors in $\mathbb{R}^n$, and consider the matrix

$$A = \begin{pmatrix} | & | & & | \\ v_1 & v_2 & \dots & v_k \\ | & | & & | \end{pmatrix}.$$

Then we can delete the columns of A without pivots (the columns corresponding to the free variables), without changing $\text{Span}\{v_1, v_2, \dots, v_k\}$.

The pivot columns are linearly independent, so we cannot delete any more columns without changing the span.

if $\{\vec{v}_1, \vec{v}_2, \dots, \vec{v}_n\}$ is linearly independent
the $A = \begin{bmatrix} | & | & & | \\ v_1 & v_2 & \dots & v_n \\ | & | & & | \end{bmatrix}$ has pivot in every column.

$$\text{Let } A = \begin{bmatrix} | & | & & | \\ v_1 & v_2 & \dots & v_n \\ | & | & & | \end{bmatrix}$$

$\{\vec{v}_1, \dots, \vec{v}_n\}$ is L.I.



pivot in every column of A



$A\vec{x} = \vec{0}$ has only the trivial solution

$\{\vec{v}_1, \dots, \vec{v}_n\}$ is L.D.



free variables exist



$A\vec{x} = \vec{0}$ has infinitely many solutions

Subspaces

Definition. A *subspace* of $\mathbb{R}^n$ is a subset V of $\mathbb{R}^n$ satisfying:

1. **Non-emptiness:** The zero vector is in V .
2. **Closure under addition:** If u and v are in V , then $u + v$ is also in V .
3. **Closure under scalar multiplication:** If v is in V and c is in $\mathbb{R}$, then cv is also in V .

Common Types of Subspaces

Theorem (Spans are Subspaces and Subspaces are Spans). If $v_1, v_2, \dots, v_p$ are any vectors in $\mathbb{R}^n$, then $\text{Span}\{v_1, v_2, \dots, v_p\}$ is a subspace of $\mathbb{R}^n$. Moreover, any subspace of $\mathbb{R}^n$ can be written as a span of a set of p linearly independent vectors in $\mathbb{R}^n$ for $p \leq n$.

Definition. Let A be an $m \times n$ matrix.

- The **column space** of A is the subspace of $\mathbb{R}^m$ spanned by the columns of A . It is written $\text{Col}(A)$.
- The **null space** of A is the subspace of $\mathbb{R}^n$ consisting of all solutions of the homogeneous equation $Ax = 0$:

$$\text{Nul}(A) = \{x \text{ in } \mathbb{R}^n \mid Ax = 0\}.$$

Basis & Dimension

Definition. Let V be a subspace of $\mathbb{R}^n$. A **basis** of V is a set of vectors $\{v_1, v_2, \dots, v_m\}$ in V such that:

1. $V = \text{Span}\{v_1, v_2, \dots, v_m\}$, and
2. the set $\{v_1, v_2, \dots, v_m\}$ is linearly independent.

Definition. Let V be a subspace of $\mathbb{R}^n$. The number of vectors in any basis of V is called the **dimension** of V , and is written $\dim V$.

Let A be an $m \times n$ matrix with r pivots.

	Col A	Nul A
Definition	All linear combinations of the columns of A	All solutions to $Ax = 0$
Subspace of	$\mathbb{R}^m$	$\mathbb{R}^n$
Dimension	$\text{rank}(A) = r = \text{number of pivots}$	$\text{nullity}(A) = n - r = \text{number of free variables}$
How to find a basis	Find the pivot positions by row reducing A ; select the corresponding columns of the original A	Solve $Ax = 0$ in parametric vector form; take the vector multiplying each free variable

$$\dim(\text{Col } A) + \dim(\text{Nul } A) = n$$

Basis Theorem. Let V be a subspace of dimension m . Then:

- Any m linearly independent vectors in V form a basis for V .
- Any m vectors that span V form a basis for V .

The Rank Theorem

Definition. The **rank** of a matrix A , written $\text{rank}(A)$, is the dimension of the column space $\text{Col}(A)$.

The **nullity** of a matrix A , written $\text{nullity}(A)$, is the dimension of the null space $\text{Nul}(A)$.

Rank Theorem. If A is a matrix with n columns, then

$$\text{rank}(A) + \text{nullity}(A) = n.$$

In other words, for any consistent system of linear equations,

$$(\text{dim of column span}) + (\text{dim of solution set}) = (\text{number of variables}).$$

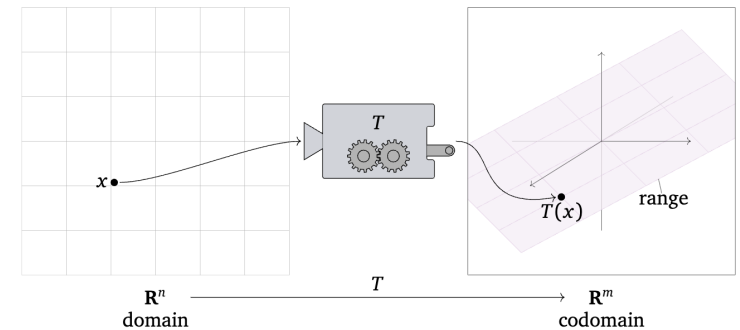
Matrix Transformations

Definition. A **transformation** from $\mathbb{R}^n$ to $\mathbb{R}^m$ is a rule T that assigns to each vector x in $\mathbb{R}^n$ a vector $T(x)$ in $\mathbb{R}^m$.

- $\mathbb{R}^n$ is called the **domain** of T .
- $\mathbb{R}^m$ is called the **codomain** of T .
- For x in $\mathbb{R}^n$, the vector $T(x)$ in $\mathbb{R}^m$ is the **image** of x under T .
- The set of all images $\{T(x) \mid x \text{ in } \mathbb{R}^n\}$ is the **range** of T .

The notation $T: \mathbb{R}^n \longrightarrow \mathbb{R}^m$ means “ T is a transformation from $\mathbb{R}^n$ to $\mathbb{R}^m$.”

It may help to think of T as a “machine” that takes x as an input, and gives you $T(x)$ as the output.



Definition. Let A be an $m \times n$ matrix. The **matrix transformation** associated to A is the transformation

$$T: \mathbf{R}^n \longrightarrow \mathbf{R}^m \quad \text{defined by} \quad T(x) = Ax.$$

This is the transformation that takes a vector x in $\mathbf{R}^n$ to the vector Ax in $\mathbf{R}^m$.

Let A be an $m \times n$ matrix, and let $T(x) = Ax$ be the associated matrix transformation.

- The **domain** of T is $\mathbf{R}^n$, where n is the number of **columns** of A .
- The **codomain** of T is $\mathbf{R}^m$, where m is the number of **rows** of A .
- The **range** of T is the **column space** of A .

